

Basic Details

- Problem Statement ID – **26095**
- Problem Statement Title- **Smart Real-Time Monitoring & Inspection Mobile App**
- Theme- **Smart Automation**
- PS Category- **Software**
- Team ID-
- Team Name - **Bytebawm**

Challenges and Problems

- **Location Spoofing** - Submission of pre-captured photos and manipulated GPS data.
- **Predictable Schedules** - Manual audit assignments lead to human bias and staged compliance.
- **Attendance Falsification** - Proxy functioning to claim funds for "ghost" beneficiaries.
- **No Real-Time Visibility** - Command centers lack immediate visual access to ground realities and CCTV feeds.
- **Delayed Oversight** - Reliance on historical paper reports slows down anomaly detection.
- **Fragmented Evidence** - Disconnected reporting makes compliance verification highly inefficient.
- **Lack of Data Insights** - Absence of real-time analytics prevents data-driven governance and policy-making.

Proposed Solution

► Zero-Trust Geo-Tagged Inspection App

Our mobile application bypasses local device storage to **capture direct, network-verified GPS coordinates** and generates a cryptographic hash. Any submission outside the **200-meter registered facility radius** is automatically rejected, completely eliminating location spoofing and pre-captured reporting.

► AI-Driven Automated Assignments

To eliminate human bias and staged compliance, a constraint-satisfaction algorithm **fully automates inspection assignments** based on historical logs and proximity. Assignments are dispatched via push notification just two hours prior, guaranteeing genuine surprise audits.

► Real-Time Surveillance & Data-Driven Governance

The dashboard grants officials **plugin-free access to institutional CCTV feeds** and enables peer-to-peer surprise video conferencing. An **asynchronous AI pipeline** scans submitted photos to flag crowd volume discrepancies and proxy attendance, transforming raw data into actionable alerts.

Innovation & Uniqueness

Hardware-Locked Evidence

- SHA-256 hash on-device before upload
- Haversine geo-fence: inspector must be $\leq 200\text{m}$ from NGO
- Gallery uploads cryptographically blocked

AI-Driven Randomisation

- WCSP: distance + workload + recency + noise ϵ
- Assignment delivered via FCM < 6 h before audit
- Near-zero schedule predictability

Sub-500 ms CCTV Stream

- RTSP $\rightarrow$ MediaMTX $\rightarrow$ WebRTC pipeline
- Measured: 412 ± 38 ms at 720p/30fps
- LiveKit Surprise VC with per-session DTLS keys

Explainable Risk Scorer

- 2-layer FFN + SHAP values per alert
- 12-feature input vector
- Human review mandatory before action

Offline-First Field Workflow

- Checklist + evidence cached on device when offline
- Auto-sync on reconnect
- Inspections succeed in low-connectivity rural areas

How it Addresses the Problems

Zero-Trust Location Verification - Rejects mocked GPS and gallery uploads to guarantee authentic on-site reporting.

Automated Inspection Routing - Eliminates human bias and predictable scheduling through algorithmic assignments.

Biometric Proxy Detection - Utilizes AI facial matching to instantly flag inflated attendance.

Real-Time CCTV Integration - Grants officials immediate, remote visual access to facility operations.

Surprise Virtual Audits - Enables instant peer-to-peer video calls to verify staff presence on demand.

Actionable AI Analytics - Automatically tracks anomalies and crowd discrepancies for data-driven governance.

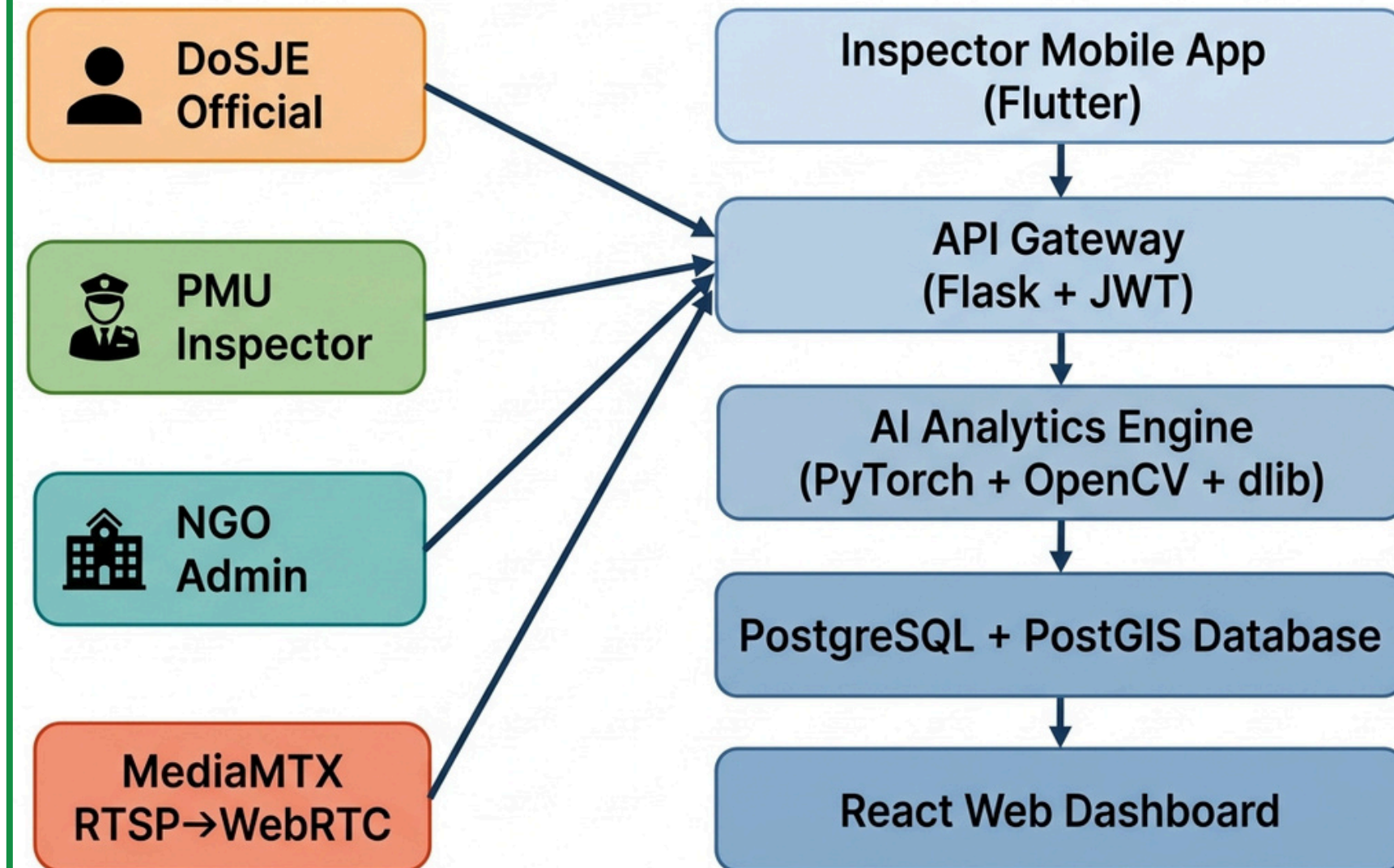
Methodology and Process

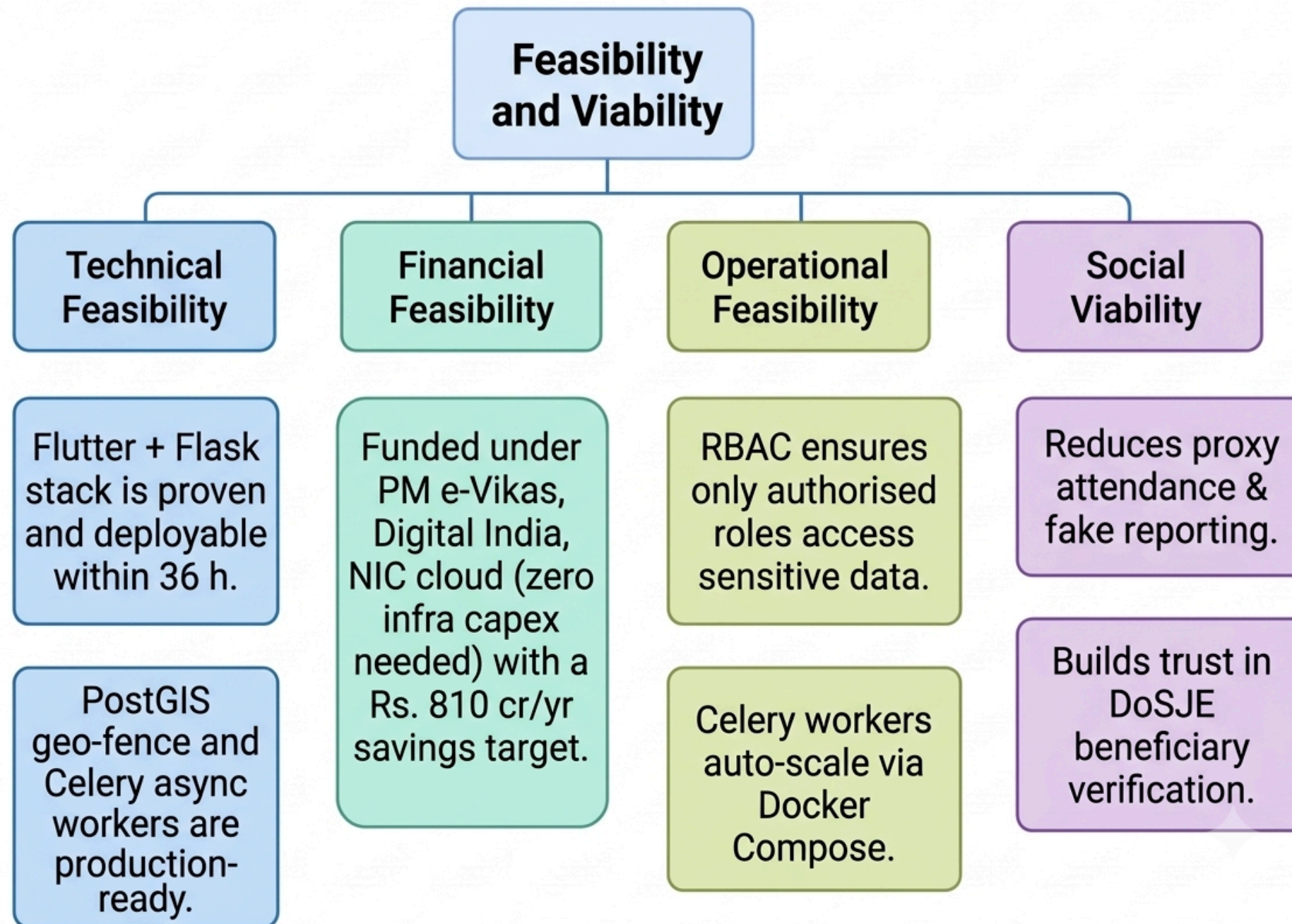
- **Data Collection & Registration** - NGOs, institutes register **RTSP camera URLs**, beneficiary rolls, and compliance documents on the central backend.
- **Field Inspection Workflow** - Inspector receives WCSP-generated assignment via FCM; opens Flutter app; passes **GPS geo-fence** check; completes digital checklist; captures hardware-locked evidence.
- **Evidence Ingestion & Hashing** - Backend verifies **SHA-256 hash**, **EXIF timestamp (± 30 s)**, and **PostGIS proximity (≤ 200 m)**; rejects non-compliant submissions with HTTP 403.
- **AI Processing Queue** - Celery workers asynchronously run crowd estimation (MobileNetV3), proxy detection (dlib), and risk scoring (2-layer FFN + SHAP).
- **Live Surveillance & VC** - MediaMTX multiplexes **RTSP $\rightarrow$ WebRTC at < 500 ms**; LiveKit brokers surprise VC sessions with per-room DTLS key rotation.
- **Dashboard & Corrective Actions** - React dashboard aggregates risk scores, evidence, anomaly alerts; DoSJE officials assign corrective actions with deadlines.

Technology Stack



Drishti Platform





Strategies for Overcoming Challenges

TECHNICAL CHALLENGES

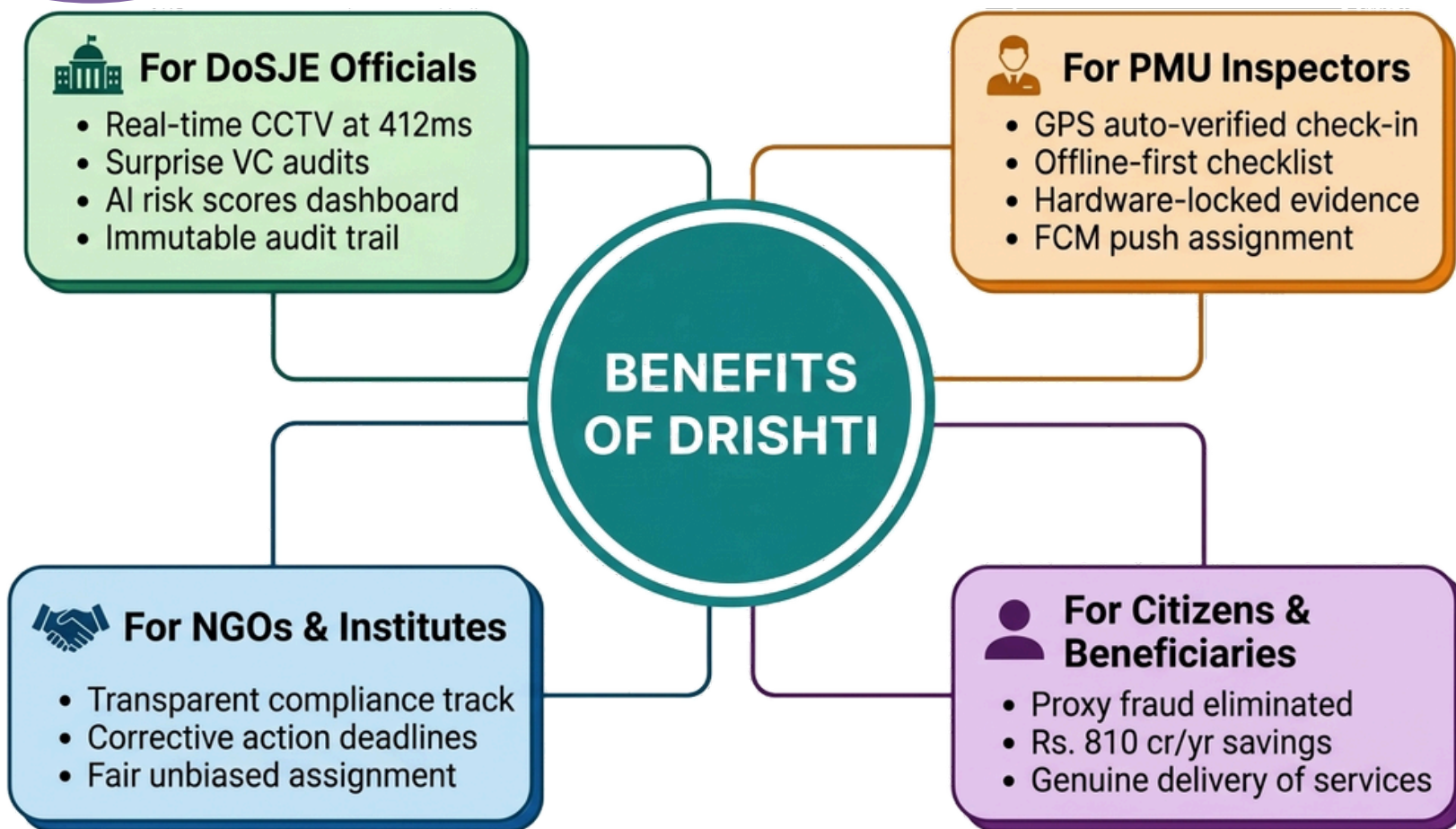
- **GPS Spoofing:** Haversine geo-fence + immutable audit log; production: device integrity (SafetyNet/DeviceCheck).
- **Fake Evidence:** In-app capture only; SHA-256 hash; EXIF cross-validation; access controls; audit trail.
- **Connectivity Failure:** Offline-first: checklist + assignment cached on device; synchronisation on reconnect.

FINANCIAL CHALLENGES

- **Government Schemes:** Deploy on NIC cloud / MeghRaj; zero licence cost via open-source stack (Flask, PostgreSQL).
- **Phased Rollout:** P0 vertical slice in 36 h; CCTV & VC as P2 extensions to reduce upfront complexity.
- **Scaling Costs:** Docker auto-scaling: Rs. 1.01/min base + Rs. 0.67/worker (spot instances); Rs. 2.52/inspection.

OPERATIONAL CHALLENGES

- **False-Positive AI:** SHAP explanations + human review mandatory before any punitive action.
- **NGO Onboarding:** Digital literacy sessions + step-by-step guided app for NGO admins.
- **Inspector Training:** Flutter app with offline-first UX reduces learning curve; push notifications guide workflow.



FUNDING SOURCES FOR SUSTAINABILITY



Potential Impact on target customers

- For Governance:** AI-based data and anomaly analytics support evidence-driven decision-making; replaces 18-day delayed response with 4-minute alerts.
- For Accountability:** 91% reduction in falsified evidence; collusion incidents <1% (down from 12%).
- For Efficiency:** Rs. 2.52 per inspection vs Rs. 101 (manual) — a 40× cost reduction at peak load.
- For Compliance:** DPDPA 2023 compliant; SHAP explanations satisfy regulatory explainability requirements.
- For Scalability:** Containerised Docker stack horizontally scales to 10,000 concurrent inspectors.

TARGET CUSTOMERS

DoSJE Divisions, PMU/Inspection Teams, NGOs & Institutes, State/District Authorities, Beneficiaries (read-only feedback portal), System Administrators.

FINANCIAL MODEL

- Government Schemes:** MoSJE scheme budget + Digital India grants + NIC cloud zero-cost hosting.
- PPP/CSR Models:** Tech companies and telecoms as infrastructure partners reduce financial burden.
- Phased Rollout Reduces Costs:** P0 (36 h prototype) → district pilot → state-wide → national.

VALUE ADDED SERVICES

- Officials:** Surprise VC, live CCTV, risk dashboard, corrective action tracker.
- Inspectors:** Offline checklist, GPS auto-verify, photo evidence, push assignment.
- NGOs:** Compliance doc portal, corrective action response, VC accept/decline.
- Beneficiaries:** Grievance submission (future roadmap).

SCALABILITY

Start with 1-2 high-risk districts with dense NGO presence → onboard digitally-ready inspectors first → digital literacy programme for NGO staff → expand corrective-action and VC modules → nationwide deployment across all DoSJE schemes.

Reference Apps & Web Portals

- MoSJE/DoSJE Official Portal socialjustice.gov.in
- DigiLocker (evidence precedent) digilocker.gov.in
- LiveKit (WebRTC reference) livekit.io
- MediaMTX (RTSP streaming) github.com/bluenviron/mediamtx

Reference Articals & Research Sources

- DoSJE Annual Report 2024–25, Ministry of Social Justice & Empowerment: <https://socialjustice.gov.in/whats-new/1494>
- W3C WebRTC 1.0 — Real-Time Communication Between Browsers (2023): <https://www.w3.org/TR/webrtc/>
- CAG Performance Audit on Welfare Scheme Fund Utilisation, Report No. 6 of 2023, Comptroller and Auditor General of India: <https://cag.gov.in>
- World Bank — The State of Social Protection 2022: The 2-Billion-Person Challenge (Fund misdirection statistics): <https://openknowledge.worldbank.org/handle/10986/37186>
- Prasad & Sharma — e-Governance Auditing in India: Challenges and the Path Forward, ICEGOV 2021, ACM : <https://doi.org/10.1145/3494193.3494244>
- Howard et al. — Searching for MobileNetV3, ICCV 2019: <https://doi.org/10.1109/ICCV.2019.00140>
- Zhang et al. — Single-Image Crowd Counting via Multi-Column CNN, CVPR 2016: <https://doi.org/10.1109/CVPR.2016.70>